Traffic Pattern Analysis

Forecasting hourly volume across anonymised monitoring stations

A slice-routed probabilistic pipeline

August 18, 2026

The problem

What we are asked to do

The deliverable

Fill `submission_template.csv`: **170,956** rows, each needing

- `forecast_volume` — point forecast
- `lower_90`, `upper_90` — 90% interval
- `reliability_score` $\in [0, 1]$

plus 5 scenario answers and a technical write-up.

How it is scored

Forecast performance	35%
Scenario analysis	20%
Uncertainty & reliability	15%
Robustness	15%
Reproducibility	10%
Traffic/technical quality	5%

Read the weights carefully

Only **35%** rewards point accuracy. **50%** rewards behaviour on hard slices, honest intervals, and scenarios. A model with slightly worse MAE but calibrated uncertainty scores higher.

The supplied data

`traffic_train.parquet`	902,902 rows
2024, 75 stations, 150 pair-series	
`traffic_validation.parquet`	556,800 rows
Jan–Jun 2025, *same* 75 stations	
`network.csv`	127 stations
lanes, speed, class, AADT band, surface	
`network_edges.csv`	186 edges
`same_corridor` / `proximity` + distance band	
`submission_template.csv`	170,956 rows

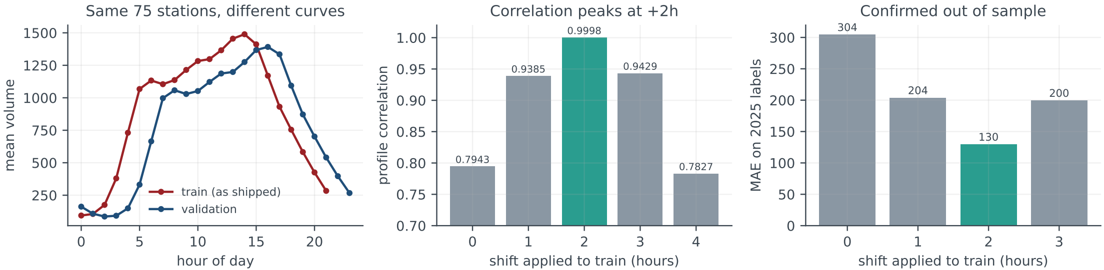
Total labelled data: 1.46M rows, ≈100 MB in memory.
This is a *structure* problem, not a big-data problem.

Two traps in the first hour

1. `volume` has *no* NaNs. Missing observations are *absent rows* — `fillna` silently does nothing.

2. `direction_code` is `str` in the parquets and `int64` in the CSVs. A naive merge returns **zero rows** and raises no error.

Finding 1 — the training clock is two hours behind



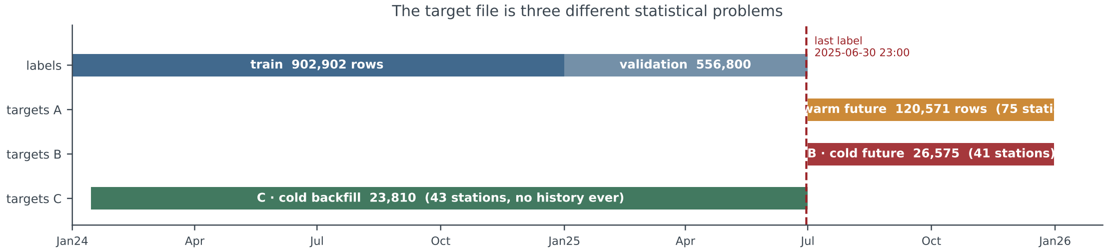
Train covers only hours 0–21, each with *exactly* 41,041 rows. Its profile peaks at 14:00 — not a real traffic curve.

Profile correlation against validation peaks at lag 2 ($r = 0.9998$), and the out-of-sample test confirms it: **MAE 304** → **MAE 130**.

Consequence

After the fix, train covers real hours 02:00–23:00 — so there is **no 2024 data at all** for target hours 00:00 and 01:00 (8.3% of rows). Lookup tables cannot fill them; cyclic features can.

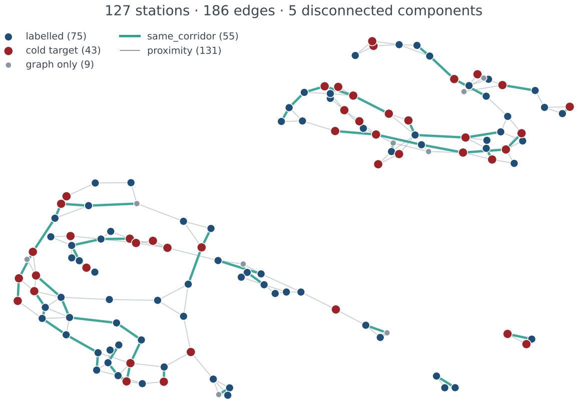
Finding 2 — the target file is three different problems



Slice	Rows	Information available at prediction time	Model class
A warm future	120,571 (70.5%)	own history to 2025-06-30, calendar, statics	direct regressio
B cold future	26,575 (15.5%)	*no* own labels, *no* network readings	inductive transfe
C cold backfill	23,810 (13.9%)	no own labels, but ∼**68 stations observed at that hour**	spatial kriging

Verified: **no target row overlaps any label**, and for all 147,146 future rows the number of stations readable at the target timestamp is **exactly zero**.

The network graph



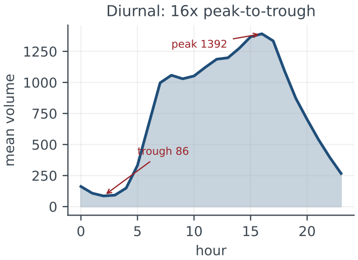
Edge types are not decorative. Residual correlation after removing each series' own profile:

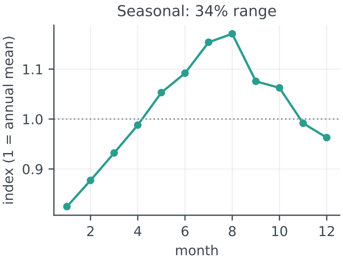
`same_corridor`	**0.844**
`proximity`	0.617
random pair	0.483

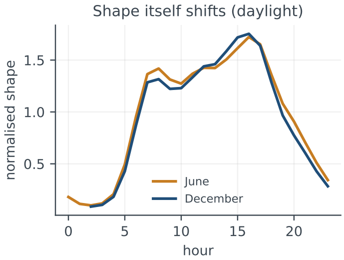
The `same_corridor` subgraph has **max degree 2** — 21 components, all simple paths. Path order is spatial order along the road, and cold stations often sit *between* labelled ones.

Even random pairs correlate 0.48: a strong network-wide common factor (weather, regional demand).

What drives the signal







Diurnal dominates: 82 veh/h at 02:00 vs 1363 at 16:00. Hour-of-day is the single most important feature.

Seasonal 34% range, and the scored horizon (Jul–Dec) spans the annual maximum *and* the descent into winter.

The shape itself moves. Evening hours run 16–21% higher in June than December — an additive month term cannot express this.

Is a six-month hourly forecast even sensible?

The objection

Predicting a specific hour six months out sounds absurd — the traffic-GNN literature (DCRNN, STGCN, STAEformer) forecasts 15–60 minutes ahead from a live sensor feed.

We measured it. Predicting May–June 2025:

Model	MAE
calendar profile, *no* recent data	**117.8**
persistence $t - 1\text{h}$ (perfect live feed)	**164.5**
persistence $t - 24\text{h}$	160.5
persistence $t - 7\text{d}$	116.7

A *perfect live sensor reading one hour before* is **worse** than knowing nothing recent but knowing the calendar.

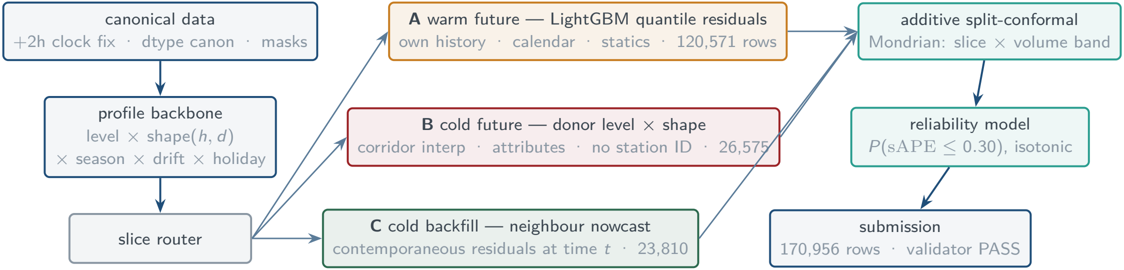
Error by month is flat: Jan 158, Feb 158, Mar 124, Apr 108, May 109, Jun 125 — it tracks *winter*, not horizon.

Verdict

This is long-term *volume forecasting* (planning), not short-term traffic prediction (operations). The task is well-posed — but the standard traffic-GNN architecture does not transfer, because its core mechanism (live neighbour readings) is unavailable for 86% of rows.

The solution

Architecture — slice-routed probabilistic pipeline



Why route by slice? A single model either discards the contemporaneous signal that exists in C, or implicitly assumes it exists in B, where it does not.

Why quantiles, not a point head? The pinball loss over $(0.05, 0.50, 0.95)$ produces the point forecast *and* the interval from one model, and absorbs heteroscedasticity directly.

The backbone — model the structure before learning residuals

Multiplicative decomposition, fitted in $\log(1+y)$ space

$$\log(1+y_{p,t}) = \underbrace{\ell_p}_{\text{level}} + \underbrace{\phi_{p,h(t),d(t)}}_{\text{hour}\times\text{dow shape}} + \underbrace{s_{m(t)}}_{\text{season}} + \underbrace{\delta_p}_{\text{2025 drift}} + \underbrace{\gamma_{\text{hol}(t)}}_{\text{holiday}} + r_{p,t}$$

Hierarchical shrinkage. The shape term is estimated at four granularities (global $\to$ axis $\to$ station $\to$ pair), each shrunk toward its parent with $w = n/(n+k)$. A thin cell falls back smoothly instead of switching hard.

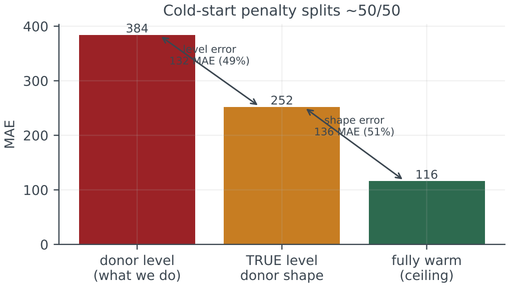
Why it matters. A 60× spread in series levels, a 16× diurnal ratio and a 34% seasonal range would all have to be rediscovered by a raw regression.

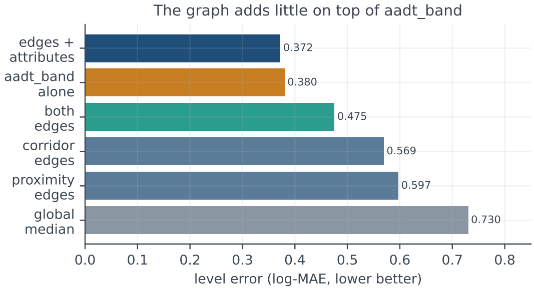
A bug worth reporting

Estimating δ_p as a fitted *slope* per pair gave slopes down to -1.5/year on outage-broken spans and extrapolated catastrophically: **MAE 173**.

Replaced with a shrunk *matched-month log ratio*: **MAE 123**.

Path B/C — transferring to stations with no history ever



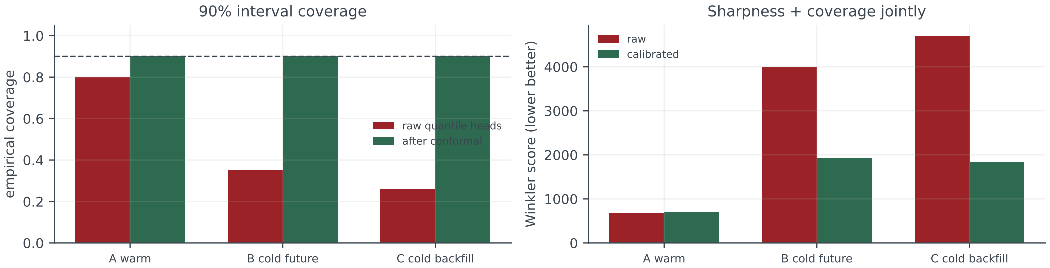


Level comes from a blend of corridor interpolation (1-D along the path), neighbour-weighted donors, and an attribute prior — then a cross-fitted LightGBM learns the combination.

An uncomfortable ablation — since overturned. `aadt_band` alone reaches log-MAE 0.380 against 0.372 for the station-level graph blend, so the graph looked worthless for level. It was the *blend* that was wrong: it averaged a station's two directions into one number. Direction-matched *pair* donors reach **0.338**, and the resulting gain is the largest on the cold slices.

Where the graph should still earn its place

Uncertainty — calibrate, do not trust the model



Raw quantile heads cover **80% / 35% / 26%** against a nominal 90% — badly overconfident exactly where the model knows least.

Split-conformal offsets fitted on strictly out-of-time, out-of-station folds restore **90%** everywhere.

Multiplicative scaling exploded

Scaling a collapsed interval to reach 90% needed a $\sim 20\times$ factor and produced **widths of 16,000** on a 622-vehicle forecast.

Additive offsets are bounded by observed residuals: Winkler **16,692** $\to$ **1,835**.

Results

Does MAE mean anything? Better metrics

GEH statistic — the domain standard

$$\mathrm{GEH} = \sqrt{\frac{2(M-C)^2}{M+C}}$$

Used by DOTs to calibrate traffic models. $\mathrm{GEH} < 5 =$ acceptable on a link; the industry target is $\geq$**85% of links below 5**.

GEH < 5, warm slice	**83.5%**
GEH < 5, cold backfill	43.6%
GEH < 5, cold future	37.2%

Metrics a planner can act on

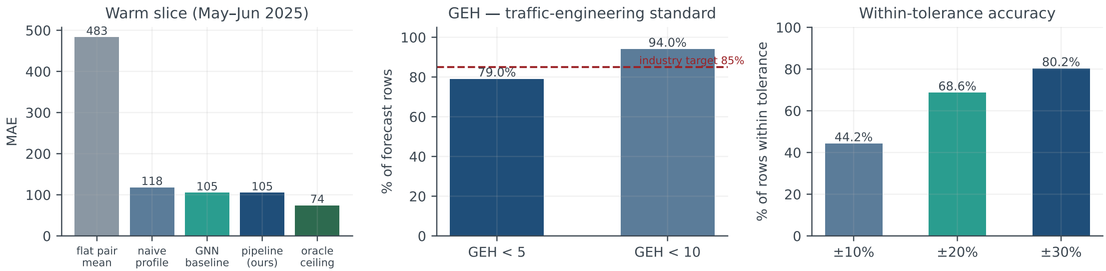
within $\pm$10% of actual	44.2% of rows
within $\pm$20%	68.6%
within $\pm$30%	80.2%
MASE (vs seasonal naive)	0.902
Skill vs naive profile	+0.109
R^2	0.958
% of achievable signal	**29%**

naive 117.8 → ours 104.9 → ceiling 74.1

The honest read

We are **just below the professional calibration standard** (79% vs 85% GEH<5) and have captured only **29%** of the gap between the naive baseline and the theoretical ceiling. Useful, not finished.

Current results



Slice	baseline MAE	**final MAE**	Winkler	coverage	rows
A warm future	116.4 backbone	**104.9**	710	0.90	120,571
B cold future	348.4 donor	336.4	1,927	0.90	26,575
C cold backfill	342.2 donor	328.8	1,835	0.90	23,810

Reliability meta-model: AUC **0.81**, Brier 0.174. Submission passes `validate_submission.py`.

A GNN baseline (relational GCN, 62k parameters, 1 s/epoch on a GTX 1050 Ti) matches the pipeline on warm MAE (104.8) but is worse calibrated (Winkler 839 vs 710).

What is left — ranked by measured expected return

1. **Hour × month interaction.** Measured: 117.8 → 109.9 MAE on the raw profile, oracle ceiling 103.0. The backbone's additive month term cannot express the seasonal shape shift. **≈8 MAE, hours of work.**
2. **Shape transfer for cold stations.** 51% of the cold penalty, and the graph barely touches it today. Transfer neighbours' normalised profile along corridors.
3. **Route by AADT reliability.** 10 of 43 cold stations have an unreliable band and 3.4× worse level error; they need the graph weighted far higher.
4. **Contemporaneous nowcast for slice C.** Currently worth only 332 → 329 MAE because the *level* is wrong; fix level first, then this signal pays.
5. **Multiple seeds and station-holdout draws.** Currently one of each — the weakest part of the validation.

Open dependency

The **5 scenario prompts** are not in the supplied data. They control **20%** of the score and no defensible answer can be produced without them.

Questions

Backup — validation design

Fold	Train	Score on
warm temporal	everything before 2025-05-01	May–Jun 2025
cold future	same cut, minus 18 whole stations	those stations, future window
cold backfill	all time, minus 18 whole stations	those stations, observed period

All folds are time-forward: the rules permit only information available at or before the predicted timestamp, so random k-fold would leak. Cold folds hold out *entire stations* — a random cell split lets the model see the held-out station's behaviour on adjacent hours and is not an honest cold-start simulation.

Cost. GNN: 62,691 parameters, ≈300 MB VRAM, 1.0 s/epoch on a GTX 1050 Ti (8.0 s on CPU). Pipeline: ≈27 LightGBM boosters, ≈4–6 GB RAM, 415 s end-to-end. *Compute is not the bottleneck; information is.*

Backup — reproducibility

`traffic_pipeline.py`	data canon, backbone, donor model, conformal, metrics
`run_pipeline.py`	folds, level model, reliability, submission
`gnn_baseline.py`	relational GCN challenger
`make_slide_figures.py`	every figure in this deck
`PROBLEM_BRIEF.md`	measured description of the dataset
`validation_results.json`	the dashboard behind these numbers

Non-negotiable canonicalisation: +2h on the training clock with calendar columns recomputed; `direction_code` cast to a single type before any join; missingness represented as absent rows, never imputed silently; the four identifier columns preserved byte-for-byte.

Environment: Python 3.12, pandas 2.2.3, LightGBM 4.6, torch 2.6.0+cu124 (the `Data-Science-Kernel` venv — the base conda env ships a CUDA build without Pascal kernels and silently falls back to CPU).